Week 8 Report — Measures & Dashboard Construction

Step 1: Creating Measures

The following measures were created in Power BI using DAX:

- **Average Math Score** → AVERAGE(StudentsPerformance[Math Score])
- **Average Reading Score** → AVERAGE(StudentsPerformance[Reading Score])
- **Average Writing Score** → AVERAGE(StudentsPerformance[Writing Score])
- **Overall Average Score** → AVERAGE(StudentsPerformance[Average Marks])
- **Overall Pass %** → DIVIDE(COUNTROWS(FILTER(PerformanceGradeTable, PerformanceGradeTable[Result] = "Pass")), COUNTROWS(PerformanceGradeTable))

👉 These measures provided the key performance indicators for the dashboard.

Step 2: Dashboard Construction

The dashboard was built in **Report View** using the prepared tables and measures. Each visual was chosen to answer specific academic questions.

- **KPI Cards**
  - Displayed Average Math, Reading, Writing, Overall Average, and Pass %.

👉 These cards give instant performance indicators at the top of the dashboard.

- **Clustered Column Chart**
  - Compared subject performance (Math vs Reading vs Writing).

👉 Helps identify the strongest and weakest subjects.

- **Clustered Bar Chart**
  - Showed parental education vs average score.

👉 Reveals how parental education influences student performance.

- **Column Chart**
    - Compared test preparation (Completed vs None) vs performance.

👉 Demonstrates the impact of preparation courses on scores.

- **Donut Chart**
    - Displayed lunch type vs performance.

👉 Highlights socioeconomic factors affecting academic outcomes.

- **Grade Distribution Chart**
    - Count of students by grade (A+, A, B+, B, C+, C, F).

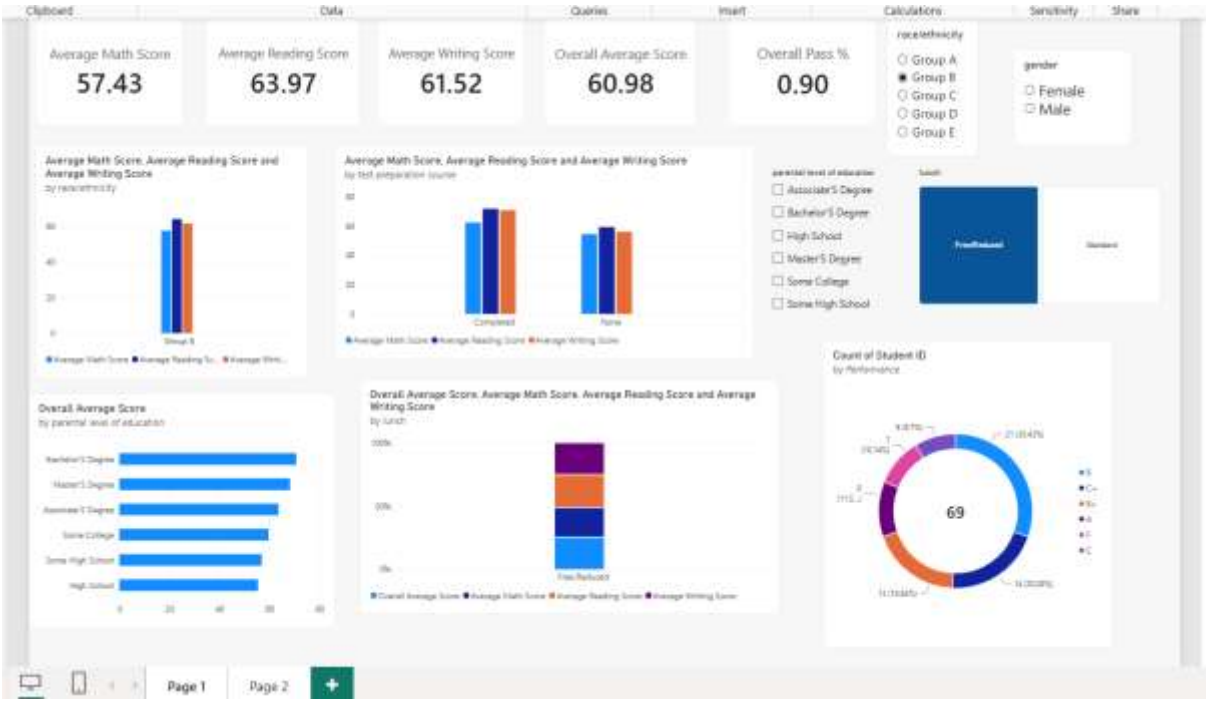
👉 Shows how students are spread across performance bands.

- **Slicers**
    - Added for Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation, Subject.

👉 Allow interactive filtering, making the dashboard dynamic and user-friendly.

- **Drill-Through Page**
    - Academic Profile page created with detailed visuals per student (Subject scores, Average Marks, Grade, Pass/Fail, Test Prep).

👉 Enables deeper analysis of individual student performance.




Report view
Overall Average Score
67.77
Visualizations
Build visual
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